

Name: _____
Class: _____
Date: _____

Mr. Rawson • GCSE Physics

RP 16: I-V Characteristics

Key concept — Ohm's law

$$V = I R \quad \text{potential difference (V) = current (A) } \times \text{ resistance } (\Omega)$$

Rearranged to find the resistance:

$$R = \frac{V}{I}$$

Two weeks ago you measured how resistance changes with the length of a nichrome wire and you got a **straight line**, showing a linear relationship. The wire followed Ohm's Law directly.

Today's question is testing if other components also follow Ohm's Law. Some components don't always follow the linear relationship between current and voltage like a wire does. As you push more current through certain devices, some devices change their resistance.

Reading an I-V graph (this is the whole skill)

You are plotting **current** on the y -axis against **potential difference** on the x -axis.

- If the I-V graph is a **Straight line** through the origin $\rightarrow$ a **linear relationship**. This shows the resistance is **constant**: doubling V doubles I .
- If the I-V graph is a **Curved line** $\rightarrow$ the component does **not** follow Ohm's law. $R = V/I$ still gives the resistance at each point, but that resistance is changing.

Test the positive AND the negative You will reverse the direction of the current flow, and take readings with the pd the other way round so you can plot points along the positive and negative axes.

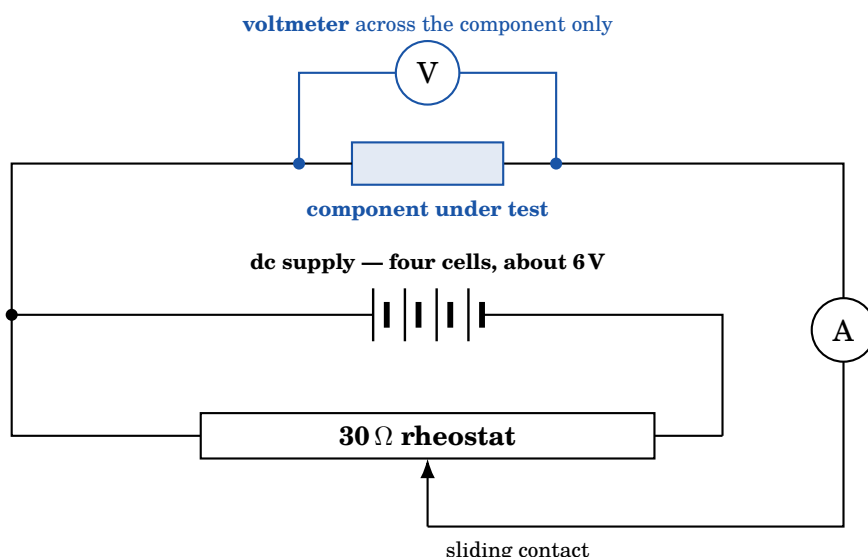
The question

How do the I-V characteristics change for different components in a circuit? Which ones follow Ohm's Law?

You will test **three components**: a fixed resistor, a filament lamp, and a diode.

Independent variable what you change	<i>The potential difference across the component — including reversing it.</i>
Dependent variable what you measure	<i>The current through the component.</i>
Control variables what stays the same	<i>The same component throughout each set of readings; the temperature of the fixed resistor — turn the supply off between readings so it does not warm up.</i>

The circuit



One circuit, three components. You build it once and swap only the blue box. The rheostat is wired with all three terminals: its two ends across the supply, and the sliding contact to the ammeter. Slide it from one end to the other to take the pd from 0 V up to the full supply. The ammeter is in the loop, so the whole current goes through it. The voltmeter goes across the component alone.

Remember your Ammeter and Voltmeter Rules

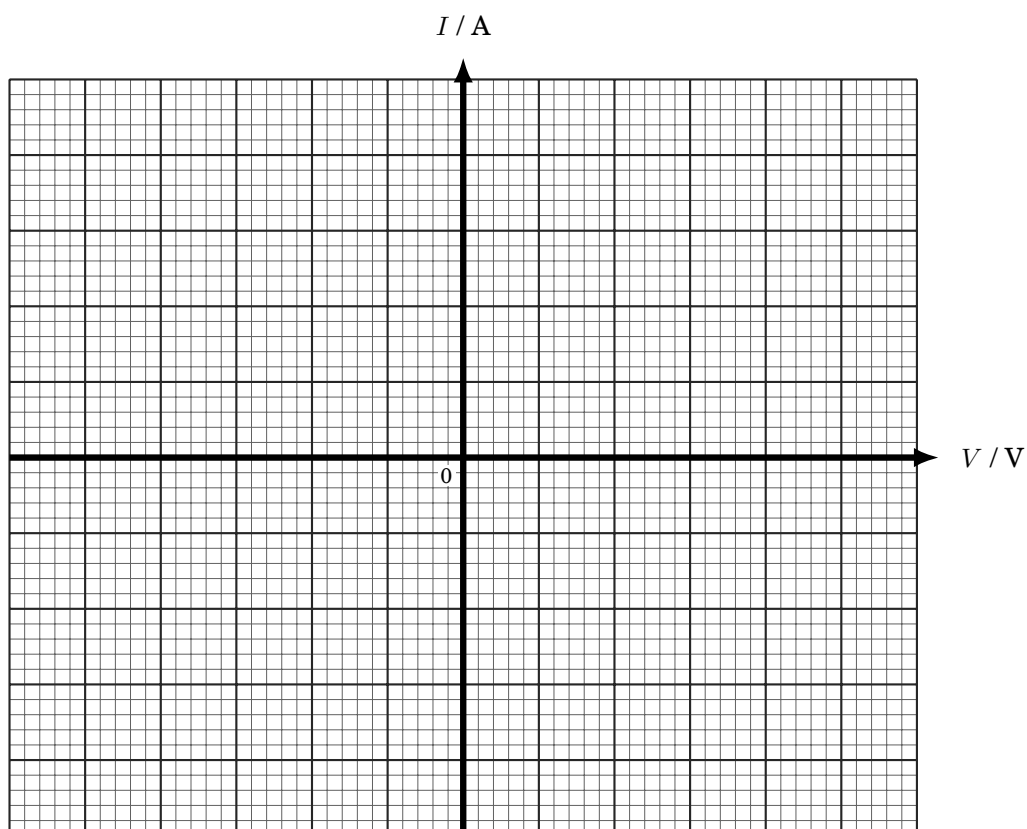
- The Ammeter must be placed **in series** with the rest of the circuit. An ideal ammeter has **zero resistance** so it does not affect the resistance in series.
- The Voltmeter must be placed **in parallel** with the rest of the circuit. An ideal voltmeter has **infinite resistance** so it will stop any current flow if it is placed in series.

Method

1. Build the circuit above with the **fixed resistor** as the component under test. Leave **one battery lead disconnected** until you have been checked.
2. Slide the rheostat all the way to the **left-hand end**, as drawn. Connect the battery lead: the voltmeter reads **0 V**. Now move the slider a short way along, read the **voltmeter** and the **ammeter** together, then **disconnect the lead again**.
3. Move the slider a little further each time and take the next pair of readings. Keep going until you reach the full supply of about **6 V**. Space your readings by the voltmeter, roughly **1 V apart**, so six or seven pairs cover the whole range. **Disconnect the lead between every reading**.
4. **Now reverse**. Disconnect, swap the two leads at the battery, and repeat the whole set. Record these readings as **negative** — both the pd and the current.
5. **Plot as you go**, straight onto the grids on the next pages. Do not wait until the end. The shape is the result, and you want to see it appearing.
6. Swap the component for the **filament lamp** and repeat steps 2–5.
7. Swap the component for the **diode** and repeat steps 2–5. Read the box on the diode page **before** you start it.

Component 1 — the fixed resistor

pd V / V	Current I / A	pd V / V	Current I / A
forward		reversed (negative)	



Label your own scale on both axes first — negative to the left of zero and below it. Then plot every point, forward and reversed, and draw **one** straight line through them all.

1) **Describe** the shape of your graph.

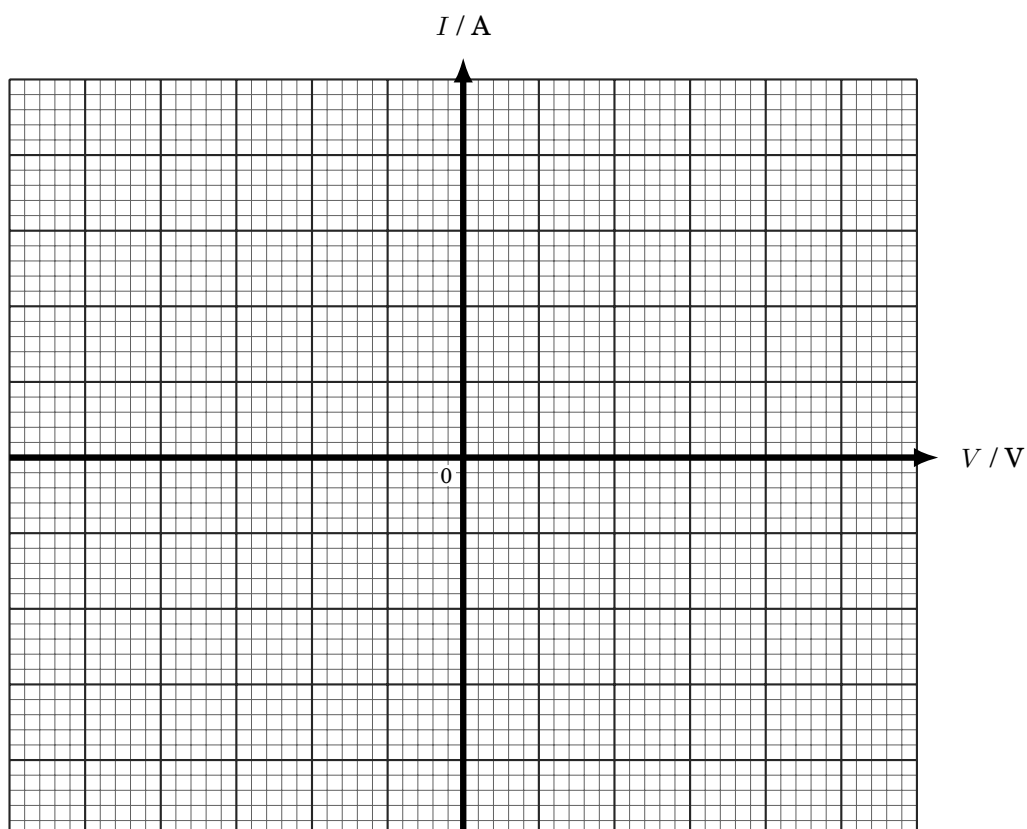
A straight line through the origin, running from bottom-left to top-right. Reversing the pd simply continues the same line into the negative quadrant.

2) **State** what that shape tells you about the resistance of the fixed resistor.

The resistance is constant. Current is directly proportional to potential difference, so V / I gives the same value at every point on the line.

Component 2 — the filament lamp

pd V / V	Current I / A	pd V / V	Current I / A
forward		reversed (negative)	



Label your own scale on both axes first. Then plot every point and draw a **smooth curve** — not a straight line, and not dot-to-dot.

3) **Describe** the shape of your graph, and say how it differs from the resistor's.

An S-shaped curve through the origin. Near the origin it is steep and almost straight, but as the pd increases in either direction the curve flattens, bending towards the horizontal axis. The resistor gave a straight line; this one does not.

4) **Explain** why the curve gets flatter as the potential difference increases. Your answer must mention temperature.

As the pd increases the current increases, and the current heats the filament. The hotter the filament, the more the metal ions vibrate, so electrons collide with them more often and the resistance increases. A larger resistance means the current rises by less for each extra volt — so the graph bends over.

5) **Explain** why the curve has the same shape on both sides of the origin.

The filament heats up whichever way the current flows — the heating does not depend on direction. So the resistance rises in the same way when the pd is reversed, giving a curve with the same shape rotated through the origin.

Component 3 — the diode

Read this before you connect the diode

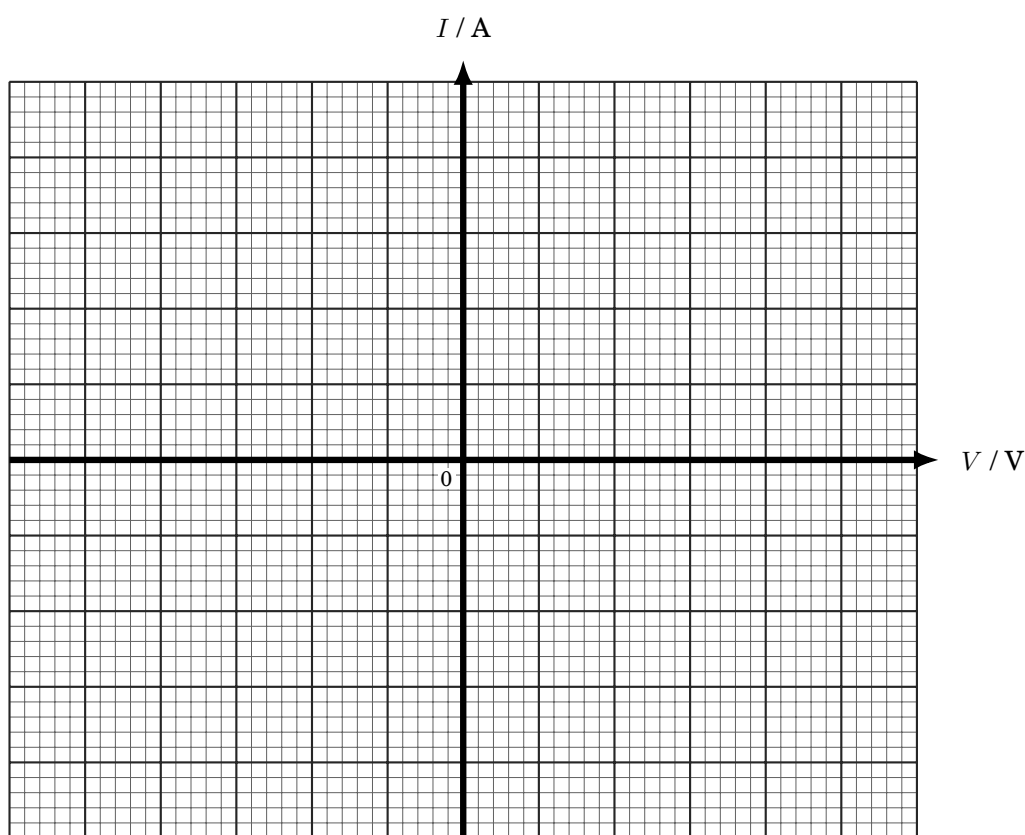
One way round, you will get a current. The other way round, you will get nothing — and nothing is the correct result.

Do not call it a broken circuit, do not go looking for a loose lead, and do not ask for a new diode. A reading of zero with several volts across it is exactly what a diode is *for*. Write the zeros down: they are your data.

Two further things you will notice in the forward direction:

- Almost nothing happens until about **0.6 V**. Below that the diode barely conducts at all.
- Once it passes that point the current climbs very steeply. **Go up in small steps here** or you will get two useful points and a mess.

pd V / V	Current I / A	pd V / V	Current I / A
forward		reversed (negative)	



Label your own scale on both axes first. The two halves of this graph do **not** match — draw what you measured, not what looks tidy.

6) **Describe** what your graph does when the potential difference is reversed.

The current stays at zero. The line lies flat along the horizontal axis for the whole of the negative side, no matter how large the reversed pd becomes.

7) **State** what this tells you about the resistance of a diode in the reverse direction.

It is extremely high — effectively infinite. A large potential difference produces no measurable current at all.

8) **Complete** the sentence. Circle one: **current through a diode flows** _____.

equally in both directions

in one direction only

only above 6 volts

only when it is hot

9) **Suggest** one job a diode could usefully do in a circuit, given what you have just measured.

Any sensible use of one-way conduction: protecting a circuit from a battery inserted the wrong way round; making sure a motor can only turn one way; converting alternating current to direct current by blocking the half of the cycle that runs backwards.

Two more components — no practical needed

Your graphs cover the three components AQA names. Two others change their resistance too, and you need to know what changes it.

Component	What changes its resistance	What it is used for
Thermistor	<i>Temperature. As the temperature rises, its resistance falls.</i>	<i>Thermostats and temperature sensors — ovens, fridges, fire alarms, car engine sensors.</i>
LDR light-dependent resistor	<i>Light intensity. As it gets brighter, its resistance falls.</i>	<i>Switching lights on at dusk — street lamps, security lights, automatic headlights, camera exposure.</i>

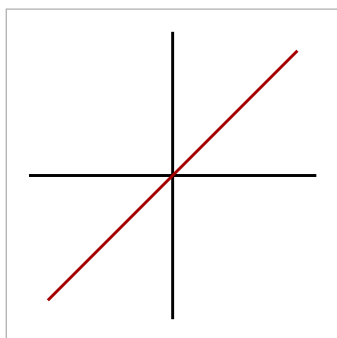
The pattern worth spotting. A fixed resistor has one resistance and keeps it. Every other component on this sheet has a resistance that *depends on something* — the current through it, its temperature, or the light falling on it.

Pulling it together

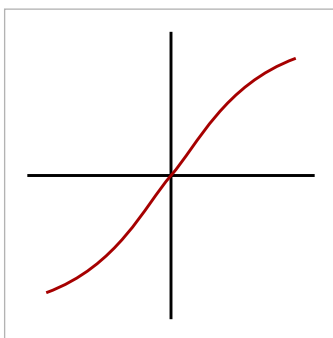
10) **Name** the one component you tested whose resistance stayed constant, and say how the graph showed it.

The fixed resistor. Its graph was a straight line through the origin, so V/I was the same at every point.

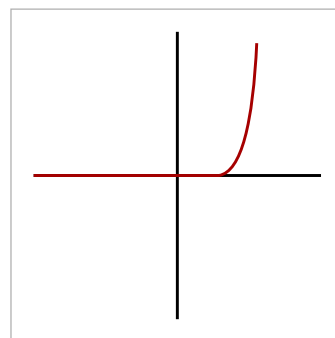
11) **Sketch** all three characteristics side by side, from memory, in the boxes below. Axes only — no numbers.



Fixed resistor



Filament lamp



Diode

Shapes only — no numbers expected. The diode's flat section must sit on the axis to the left of the origin, and the lamp's curve must be symmetrical about the origin.

Evaluation

12) **Explain** why the readings for the fixed resistor would be wrong if you left the supply on for the whole experiment.

The current would warm the resistor. A warmer resistor has a greater resistance, so the later readings would give a smaller current than they should and the line would bend instead of staying straight — and the component would no longer be “at constant temperature”, which is the condition the experiment is meant to test.

13) **Give** one reason why the filament lamp's graph is a curve while the fixed resistor's is a straight line, even though both are just pieces of metal wire.

The filament gets far hotter. It is a very thin wire designed to glow, so its temperature changes enormously over the range of pd used, and resistance rises with temperature. The fixed resistor barely warms at all, so its resistance stays effectively constant.

14) **Suggest** one improvement to the method that would make the diode's turn-on point easier to identify.

Take many more readings in small steps between about 0.4 V and 0.8 V, rather than using the same large steps as the rest of the range — and repeat them to take a mean. The turn-on happens over a very narrow range of pd, so widely spaced readings step straight over it.

Extra space for working

A second attempt at a graph, a repeated set of readings, anything that did not fit.

Before you pack away: disconnect the battery, and check the lamp is *cool* before it goes back in the tray. Diodes back in their own box — they are easy to lose and easy to destroy. Meters back the way you found them.